

Topic 09: Laplace Transform of Integrals

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

Topic 08 showed you something remarkable: differentiation in the t -domain becomes multiplication by s in the s -domain. Now here is the perfect companion result: integration in the t -domain becomes division by s . So the entire calculus — differentiation and integration — maps to simple arithmetic in the s -domain. Multiply for derivatives, divide for integrals. This topic proves the division result and shows you how to use it to handle integral equations and accumulation problems.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this... Integral equations appear in every branch of engineering. In electrical engineering the charge accumulated on a capacitor is $q(t) = \int_0^t i(\tau) d\tau$, where $i(\tau)$ is the current. In mechanics the displacement of a particle is the time integral of its velocity. In control systems the integral of an error signal drives an integral controller. Without the Laplace transform of an integral you cannot solve any of these problems algebraically.

Mistake	Consequence
Not knowing the integral transform formula	Unable to solve integral equations using Laplace methods
Forgetting to divide by s (only multiplying)	Getting the derivative formula instead of the integral formula
Confusing $\mathcal{L}\left\{\int_0^t f\right\}$ with $\mathcal{L}\{f\}/s$ as a general rule	Only correct if lower limit is 0 — wrong otherwise

Key Takeaway

Integrals in the t -domain become division by s in the s -domain, provided the lower limit of integration is 0.

3 You Already Know This (Intuition First)

Think about the relationship between multiplication and division in ordinary arithmetic. When you know that differentiation corresponds to multiplication by s , it is completely natural to expect that integration — the inverse operation — corresponds to division by s . Just as the exponential and the logarithm are inverses (one “goes up” a level, the other “comes down”), differentiation and integration in the t -domain are inverses of each other, and they correspond to multiplication and division by s respectively.

Here is another way to feel this. Integration is an *accumulation* operation: the integral $\int_0^t f(\tau) d\tau$ is the running total of f up to time t . In the s -domain that accumulation over an extra variable corresponds to one more factor of $1/s$ — exactly the same way that $\int_0^\infty e^{-st} dt = 1/s$ introduces a factor of $1/s$ for each integration over time. This is not a coincidence; it is the essence of the transform.

4 The Main Theorem — Transform of an Integral

Theorem: Laplace Transform of an Integral

Theorem. Let $f(t)$ be a piecewise continuous function of exponential order, and let $\mathcal{L}\{f(t)\} = F(s)$. Then

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}, \quad s > 0.$$

Proof — Method 1 (using the derivative formula from Topic 08).

Define $g(t) = \int_0^t f(\tau) d\tau$. By the Fundamental Theorem of Calculus, $g'(t) = f(t)$.

Also note $g(0) = \int_0^0 f(\tau) d\tau = 0$.

By the Laplace derivative formula (Topic 08):

$$\mathcal{L}\{g'(t)\} = sG(s) - g(0) = sG(s) - 0 = sG(s).$$

But $g'(t) = f(t)$, so $\mathcal{L}\{g'(t)\} = \mathcal{L}\{f(t)\} = F(s)$.

Therefore $F(s) = sG(s)$, which gives

$$G(s) = \frac{F(s)}{s}, \quad \text{i.e.,} \quad \mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}. \quad \square$$

Proof — Method 2 (direct — change of order of integration).

Starting from the definition:

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \int_0^\infty e^{-st} \left[\int_0^t f(\tau) d\tau \right] dt.$$

The region of integration in the (τ, t) -plane is $\{0 \leq \tau \leq t < \infty\}$. Reversing the order of integration (holding τ fixed and integrating t from τ to ∞ first, then τ from 0 to ∞):

$$= \int_0^\infty f(\tau) \left[\int_\tau^\infty e^{-st} dt \right] d\tau.$$

Evaluate the inner integral:

$$\int_\tau^\infty e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_\tau^\infty = 0 - \frac{e^{-s\tau}}{-s} = \frac{e^{-s\tau}}{s}.$$

Substituting back:

$$= \int_0^\infty f(\tau) \cdot \frac{e^{-s\tau}}{s} d\tau = \frac{1}{s} \int_0^\infty e^{-s\tau} f(\tau) d\tau = \frac{F(s)}{s}. \quad \square$$

What Did We Learn?

Both proofs confirm $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = F(s)/s$. Method 1 is slick (one line using the derivative formula). Method 2 is foundational (shows the geometry of interchanging integration order).

5 Inverse Form

Inverse Form of the Integral Theorem

Taking the inverse Laplace transform of both sides of the main theorem:

$$\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(\tau) d\tau,$$

where $f(t) = \mathcal{L}^{-1}\{F(s)\}$.

Interpretation. Whenever you see a transform of the form $F(s)/s$ (i.e., $F(s)$ multiplied by $1/s$), the inverse transform is the running integral of $f(t)$ from 0 to t . The “bonus $1/s$ ” means: integrate once.

Worked Example. Find $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$.

Write $\frac{1}{s^3} = \frac{1/s^2}{s}$. We know $\mathcal{L}^{-1}\{1/s^2\} = t$. By the inverse form, the extra $1/s$ means integrate:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \int_0^t \tau d\tau = \left[\frac{\tau^2}{2}\right]_0^t = \frac{t^2}{2}.$$

Verification. $\mathcal{L}\left\{\frac{t^2}{2}\right\} = \frac{1}{2} \cdot \frac{2!}{s^3} = \frac{1}{s^3}$. ✓

What Did We Learn?

A factor of $1/s$ in the s -domain corresponds to one integration in the t -domain. Use this to invert transforms that have an “extra” $1/s$ factor.

6 Repeated Integration

Repeated Integration Formula

Theorem. If $\mathcal{L}\{f(t)\} = F(s)$, then applying the integral theorem n times:

$$\mathcal{L}\left\{\underbrace{\int_0^t \int_0^{t_1} \cdots \int_0^{t_{n-1}}}_{n \text{ times}} f(t_n) dt_n \cdots dt_1\right\} = \frac{F(s)}{s^n}.$$

Case $n = 2$ (explicit derivation). Let $g(t) = \int_0^t f(\tau) d\tau$. By the single-integral

theorem:

$$\mathcal{L}\{g(t)\} = \frac{F(s)}{s}.$$

Now define $h(t) = \int_0^t g(u) du$. Applying the theorem again with g in place of f and $G(s) = F(s)/s$ in place of $F(s)$:

$$\mathcal{L}\{h(t)\} = \frac{G(s)}{s} = \frac{F(s)/s}{s} = \frac{F(s)}{s^2}.$$

But $h(t) = \int_0^t \int_0^{t_1} f(\tau) d\tau dt_1$, so:

$$\mathcal{L}\left\{\int_0^t \int_0^{t_1} f(\tau) d\tau dt_1\right\} = \frac{F(s)}{s^2}.$$

Each additional integration multiplies the denominator by one more factor of s .

What Did We Learn?

n nested integrations from 0 to t divide the transform by s^n . Each “layer” of integration adds one factor of $1/s$ in the s -domain.

7 Application — Integral Equations

Solving Integral Equations with Laplace Transforms

Background. A Volterra integral equation involves an unknown function $f(t)$ under an integral sign. The Laplace transform converts such equations to algebraic equations in $F(s)$, which can be solved directly.

Volterra Equation of the First Kind. Given: $\int_0^t f(\tau) d\tau = g(t)$.

Take $\mathcal{L}$ of both sides:

$$\frac{F(s)}{s} = G(s) \implies F(s) = s G(s) \implies f(t) = g'(t).$$

The integral equation is solved by differentiation — an elegant result.

Volterra Equation of the Second Kind. Solve: $f(t) + \int_0^t f(\tau) d\tau = 1$.

Step 1. Take $\mathcal{L}$ of both sides:

$$F(s) + \frac{F(s)}{s} = \frac{1}{s}.$$

Step 2. Factor out $F(s)$:

$$F(s) \left(1 + \frac{1}{s}\right) = \frac{1}{s} \implies F(s) \cdot \frac{s+1}{s} = \frac{1}{s}.$$

Step 3. Solve for $F(s)$:

$$F(s) = \frac{1}{s} \cdot \frac{s}{s+1} = \frac{1}{s+1}.$$

Step 4. Invert:

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}.$$

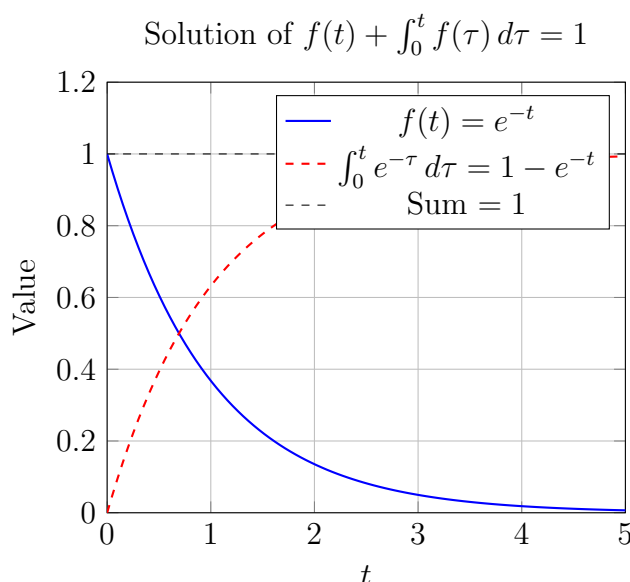
Verification. $f(t) = e^{-t}$, so $\int_0^t e^{-\tau} d\tau = 1 - e^{-t}$. Then $f(t) + \int_0^t f(\tau) d\tau = e^{-t} + (1 - e^{-t}) = 1$. ✓

What Did We Learn?

The Laplace transform converts a Volterra integral equation into a simple algebraic equation. For the first kind: $F(s) = sG(s)$, so $f = g'$. For the second kind: factor out $F(s)$ and solve.

Graphical Illustration

Below is a plot of the solution $f(t) = e^{-t}$ and its running integral $\int_0^t e^{-\tau} d\tau = 1 - e^{-t}$, which together sum to 1, confirming the integral equation.



8 Worked Examples

Example 1: $\mathcal{L}\left\{\int_0^t e^{-2\tau} d\tau\right\}$

Method A — Using the theorem.

We know $\mathcal{L}\{e^{-2t}\} = \frac{1}{s+2}$.

By the integral theorem:

$$\mathcal{L}\left\{\int_0^t e^{-2\tau} d\tau\right\} = \frac{1}{s(s+2)}.$$

Method B — Direct evaluation then transform.

First evaluate the integral:

$$\int_0^t e^{-2\tau} d\tau = \left[\frac{e^{-2\tau}}{-2} \right]_0^t = \frac{1 - e^{-2t}}{2}.$$

Now take the transform:

$$\mathcal{L}\left\{\frac{1 - e^{-2t}}{2}\right\} = \frac{1}{2} \left(\frac{1}{s} - \frac{1}{s+2} \right) = \frac{1}{2} \cdot \frac{(s+2) - s}{s(s+2)} = \frac{1}{2} \cdot \frac{2}{s(s+2)} = \frac{1}{s(s+2)}.$$

Both methods give the same answer. ✓

What Did We Learn? (Example 1)

The theorem $\mathcal{L}\left\{\int_0^t f\right\} = F(s)/s$ avoids computing the integral explicitly. But direct evaluation is a powerful cross-check.

Example 2: $\mathcal{L}\left\{\int_0^t \sin(3\tau) d\tau\right\}$

Step 1. Find $F(s)$:

$$\mathcal{L}\{\sin(3t)\} = \frac{3}{s^2 + 9}.$$

Step 2. Apply the integral theorem:

$$\mathcal{L}\left\{\int_0^t \sin(3\tau) d\tau\right\} = \frac{F(s)}{s} = \frac{3}{s(s^2 + 9)}.$$

Verification. Evaluate the integral directly:

$$\int_0^t \sin(3\tau) d\tau = \left[\frac{-\cos(3\tau)}{3} \right]_0^t = \frac{1 - \cos(3t)}{3}.$$

Take its transform:

$$\mathcal{L}\left\{\frac{1 - \cos(3t)}{3}\right\} = \frac{1}{3} \left(\frac{1}{s} - \frac{s}{s^2 + 9} \right) = \frac{1}{3} \cdot \frac{(s^2 + 9) - s^2}{s(s^2 + 9)} = \frac{1}{3} \cdot \frac{9}{s(s^2 + 9)} = \frac{3}{s(s^2 + 9)}.$$

Both methods agree. ✓

What Did We Learn? (Example 2)

For a sine function, the theorem gives $F(s)/s = 3/[s(s^2 + 9)]$. Splitting $1 - \cos(3t)$ into two transforms and combining confirms the result.

Example 3: Solve $y(t) = t - \int_0^t y(\tau) d\tau$

This is a Volterra integral equation of the second kind with $g(t) = t$.

Step 1. Take $\mathcal{L}$ of both sides:

$$Y(s) = \frac{1}{s^2} - \frac{Y(s)}{s}.$$

Step 2. Collect $Y(s)$ terms on the left:

$$Y(s) + \frac{Y(s)}{s} = \frac{1}{s^2} \implies Y(s) \left(\frac{s+1}{s} \right) = \frac{1}{s^2}.$$

Step 3. Solve for $Y(s)$:

$$Y(s) = \frac{1}{s^2} \cdot \frac{s}{s+1} = \frac{1}{s(s+1)}.$$

Step 4. Apply partial fractions:

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}.$$

Step 5. Invert:

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{1}{s+1} \right\} = 1 - e^{-t}.$$

Verification. $y(t) = 1 - e^{-t}$ and $\int_0^t y(\tau) d\tau = \int_0^t (1 - e^{-\tau}) d\tau = [\tau + e^{-\tau}]_0^t = (t + e^{-t}) - (0 + 1) = t + e^{-t} - 1$. Then:

$$t - \int_0^t y(\tau) d\tau = t - (t + e^{-t} - 1) = 1 - e^{-t} = y(t).$$

Matches $y(t)$. ✓

What Did We Learn? (Example 3)

The Laplace method transforms the integral equation into algebra in three steps: transform, collect, invert. The result $y = 1 - e^{-t}$ grows from 0 and approaches 1, consistent with the accumulating integral on the right-hand side.

9 Duality Summary Table

The table below summarises how each calculus operation in the t -domain maps to an algebraic operation in the s -domain. Topics 10 and 11 complete the picture for multiplication/division by t .

Operation in t -domain	Effect in s -domain	Topic
Differentiation: $f'(t)$	Multiplication by s : $sF(s) - f(0)$	Topic 08
Integration: $\int_0^t f(\tau) d\tau$	Division by s : $F(s)/s$	Topic 09
Multiplication by t : $t f(t)$	Differentiation: $-F'(s)$	Topic 10
Division by t : $f(t)/t$	Integration: $\int_s^\infty F(u) du$	Topic 11

10 Excel Example

We numerically verify $\mathcal{L}\left\{\int_0^t e^{-\tau} d\tau\right\}\Big|_{s=2}$.

The integral is $g(t) = 1 - e^{-t}$, so $\mathcal{L}\{g\} = \frac{1}{s} - \frac{1}{s+1} = \frac{1}{s(s+1)}$. At $s = 2$: $\frac{1}{2 \times 3} = \frac{1}{6} \approx 0.16667$.

Spreadsheet layout (columns A–D, rows 2 onwards, step 0.1 from $t = 0$ to $t = 6$):

Row	Column A t	Column B $g(t) = 1 - e^{-t}$	Column C $e^{-2t} g(t)$	Column D Cumul. Trapezoid Sum
1	t	g(t)	$e^{-2t} g(t)$	Trap. Sum
2	0	=1-EXP(-A2)	=EXP(-2*A2)*B2	0
3	=A2+0.1	=1-EXP(-A3)	=EXP(-2*A3)*B3	=D2+0.5*(C2+C3)*0.1
...	(drag down to row 62, i.e. $t = 6$)			

Explicit cell formulas:

- **B2:** =1-EXP(-A2) ($g(t)$ at $t = 0$ gives 0)
- **C2:** =EXP(-2*A2)*B2 (integrand $e^{-2t}(1 - e^{-t})$)
- **D3:** =D2+0.5*(C2+C3)*0.1 (trapezoid rule with $\Delta t = 0.1$)
- In row 62 ($t = 6$): $e^{-12}(1 - e^{-6}) \approx 0$ so the cumulative sum has converged.

Sample computed rows:

t	$g(t) = 1 - e^{-t}$	$e^{-2t} g(t)$	Cumul. Trap. Sum
0.0	0.00000	0.00000	0.00000
0.5	0.39347	0.14475	0.04928
1.0	0.63212	0.08556	0.10891
2.0	0.86466	0.01584	0.15296
3.0	0.95021	0.00236	0.16444
6.0	0.99752	0.00001	0.16658

What Did We Learn? (Excel)

The trapezoid sum converges to ≈ 0.16658 , very close to the exact value $1/6 = 0.16667$. The small discrepancy is due to the finite step size $\Delta t = 0.1$ and the truncation at $t = 6$. A smaller step or larger upper limit improves accuracy.

11 Python Example

Listing 1: Laplace Transform of Integrals — Symbolic Verification

```

1 import sympy as sp
2
3 t, tau, s = sp.symbols('t tau s', positive=True)
4
5 # --- Symbolic computation of  $L\{\int_0^t f(\tau) d\tau\}$  ---
6 def laplace_of_integral(f_tau):
7     """Return  $F(s)/s$  where  $F(s) = L\{f(t)\}$ ."""
8     F_s = sp.laplace_transform(f_tau.subs(tau, t), t, s)[0]
9     return sp.simplify(F_s / s)
10
11 # Test functions
12 cases = {
13     'sin(t)': sp.sin(tau),
14     't^2': tau**2,
15     'e^{-3t}': sp.exp(-3*tau),
16 }
17
18 print("=" * 55)
19 print("Verification:  $L\{\int_0^t f(\tau) d\tau\} = F(s)/s$ ")
20 print("=" * 55)
21 for name, f_tau in cases.items():
22     # Compute integral  $\int_0^t$  then take Laplace transform
23     integral_f = sp.integrate(f_tau, (tau, 0, t))
24     lhs = sp.laplace_transform(integral_f, t, s)[0]
25     lhs = sp.simplify(lhs)
26     rhs = laplace_of_integral(f_tau)
27     match = sp.simplify(lhs - rhs) == 0
28     print(f"\nf(t) = {name}")
29     print(f"  L{\int_0^t f} = {lhs}")
30     print(f"  F(s)/s      = {rhs}")
31     print(f"  Match        : {match}")
32
33 # Expected output:
34 # =====
35 # Verification:  $L\{\int_0^t f(\tau) d\tau\} = F(s)/s$ 
36 # =====
37 #
38 # f(t) = sin(t)
39 #   L{\int_0^t f} = 1/(s*(s**2 + 1))
40 #   F(s)/s      = 1/(s*(s**2 + 1))

```

```

41 # Match : True
42 #
43 # f(t) = t^2
44 # L{int_0^t f} = 2/s**4
45 # F(s)/s = 2/s**4
46 # Match : True
47 #
48 # f(t) = e^{-3t}
49 # L{int_0^t f} = 1/(s*(s + 3))
50 # F(s)/s = 1/(s*(s + 3))
51 # Match : True
52
53 # --- Solve Example 3: y(t) = t - int_0^t y(tau) dtau ---
54 print("\n" + "=" * 55)
55 print("Solving: y(t) = t - int_0^t y(tau) dtau")
56 print("=" * 55)
57 Y = sp.Function('Y')
58 y = sp.Function('y')
59
60 # Algebraic approach in s-domain
61 Ys = sp.Symbol('Ys')
62 # Y(s) = 1/s^2 - Y(s)/s => Y(s)(1 + 1/s) = 1/s^2
63 eq = sp.Eq(Ys * (1 + 1/s), 1/s**2)
64 sol_Ys = sp.solve(eq, Ys)[0]
65 sol_Ys = sp.simplify(sol_Ys)
66 print(f"Y(s) = {sol_Ys}")
67
68 # Partial fractions
69 pf = sp.apart(sol_Ys, s)
70 print(f"Partial fractions: {pf}")
71
72 # Inverse Laplace
73 y_t = sp.inverse_laplace_transform(sol_Ys, s, t)
74 y_t = sp.simplify(y_t)
75 print(f"y(t) = {y_t}")
76
77 # Expected output:
78 # =====
79 # Solving: y(t) = t - int_0^t y(tau) dtau
80 # =====
81 # Y(s) = 1/(s*(s + 1))
82 # Partial fractions: 1/s - 1/(s + 1)
83 # y(t) = 1 - exp(-t)

```

What Did We Learn? (Python)

sympy confirms all three integral transforms equal $F(s)/s$. The symbolic solver for Example 3 returns $Y(s) = 1/[s(s + 1)]$ and $y(t) = 1 - e^{-t}$, matching our hand calculation exactly.

12 Viva-Style Oral Questions

Q1. State the Laplace transform of an integral.

Answer. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$, provided $s > 0$ and f is piecewise continuous of exponential order.

Q2. Prove the integral transform formula using the derivative formula (Method 1).

Answer. Let $g(t) = \int_0^t f(\tau) d\tau$. Then $g'(t) = f(t)$ and $g(0) = 0$. By the first derivative formula: $\mathcal{L}\{g'\} = sG(s) - g(0) = sG(s)$. Since $\mathcal{L}\{g'\} = F(s)$, we get $sG(s) = F(s)$, so $G(s) = F(s)/s$.

Q3. Prove the integral transform formula by changing the order of integration (Method 2).

Answer. Write $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \int_0^\infty e^{-st} \int_0^t f(\tau) d\tau dt$. The region $0 \leq \tau \leq t < \infty$ is reversed to $\tau \leq t < \infty, 0 \leq \tau < \infty$. The inner integral becomes $\int_\tau^\infty e^{-st} dt = e^{-s\tau}/s$. Substituting gives $(1/s) \int_0^\infty e^{-s\tau} f(\tau) d\tau = F(s)/s$.

Q4. What is the inverse form of the integral theorem?

Answer. $\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(\tau) d\tau$, where $f(t) = \mathcal{L}^{-1}\{F(s)\}$. A factor of $1/s$ in the s -domain corresponds to one integration from 0 to t in the t -domain.

Q5. What does $\mathcal{L}^{-1}\{F(s)/s\}$ give?

Answer. It gives the running integral of $f(t)$ from 0 to t , i.e., $\int_0^t f(\tau) d\tau$. For example, $\mathcal{L}^{-1}\{1/s^3\} = \mathcal{L}^{-1}\{(1/s^2)/s\} = \int_0^t \tau d\tau = t^2/2$.

Q6. State and derive the repeated integration formula.

Answer. $\mathcal{L}\left\{\int_0^t \int_0^{t_1} \cdots \int_0^{t_{n-1}} f(t_n) dt_n \cdots dt_1\right\} = F(s)/s^n$. Derived by applying the single-integral formula n times. At each step the denominator gains one more factor of s .

Q7. How do you solve a Volterra integral equation using Laplace transforms?

Answer. Take $\mathcal{L}$ of both sides. The integral becomes $F(s)/s$, converting the equation to an algebraic equation in $F(s)$. Solve for $F(s)$ algebraically, then invert to find $f(t)$. For example, $f + \int_0^t f d\tau = 1$ gives $F(1 + 1/s) = 1/s \Rightarrow F = 1/(s + 1) \Rightarrow f = e^{-t}$.

Q8. Explain the derivative–integral duality in the Laplace transform.

Answer. In the t -domain, differentiation and integration are inverse operations. In the s -domain they correspond to multiplication and division by s respectively. More precisely: $\mathcal{L}\{f'\} = sF(s) - f(0)$ (multiply by s , subtract initial condition); and $\mathcal{L}\left\{\int_0^t f\right\} = F(s)/s$ (divide by s). This duality converts all of calculus into algebra.

13 Descriptive Questions

1. Prove the formula $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = F(s)/s$ using both the derivative-formula method and the change-of-order method. State clearly what conditions on f are required.
2. Find $\mathcal{L}\left\{\int_0^t \tau^2 d\tau\right\}$ step by step using the integral theorem. Verify by evaluating the integral explicitly and then taking its Laplace transform.
3. Solve the Volterra integral equation $y(t) + 2\int_0^t y(\tau) d\tau = \sin t$ using Laplace transforms. Show every step from applying $\mathcal{L}$ through inverting $Y(s)$.
4. Using the inverse form of the integral theorem, find $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$. Verify your answer by computing $\mathcal{L}$ of the result.
5. Explain the duality between the Laplace transforms of derivatives and integrals. How does this duality turn every calculus problem in the t -domain into an algebraic problem in the s -domain? Give one example for each operation.

14 Practice Problems

P1. Find $\mathcal{L}\left\{\int_0^t \cos(2\tau) d\tau\right\}$. *Hint:* $\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2 + 4}$; then $\frac{F(s)}{s} = \frac{1}{s^2 + 4}$. *Answer:* $\frac{1}{s^2 + 4}$.

P2. Find $\mathcal{L}\left\{\int_0^t \tau e^{-\tau} d\tau\right\}$. *Hint:* $\mathcal{L}\{te^{-t}\} = 1/(s+1)^2$; *answer:* $\frac{1}{s(s+1)^2}$.

P3. Find $\mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\}$. *Hint:* $2/s^3 = (2/s^2)/s$; $\mathcal{L}^{-1}\{2/s^2\} = 2t$; *integrate:* $\int_0^t 2\tau d\tau = t^2$. *Answer:* t^2 .

P4. Solve the integral equation $y(t) = \sin t - \int_0^t y(\tau) d\tau$. *Hint:* $Y(s)(1+1/s) = 1/(s^2+1)$; $Y(s) = s/[(s+1)(s^2+1)]$; *partial fractions give* $y(t) = -\frac{1}{2}e^{-t} + \frac{1}{2}\sin t + \frac{1}{2}\cos t$.

P5. Find $\mathcal{L}\left\{\int_0^t \int_0^{t_1} e^{-\tau} d\tau dt_1\right\}$. *Hint:* *repeated integration;* $\mathcal{L}\{e^{-t}\} = 1/(s+1)$; *answer:* $\frac{1}{s^2(s+1)}$.

P6. Given $\mathcal{L}\{f\} = \frac{3}{s^2(s+1)}$, find $f(t)$. *Hint:* *partial fractions:* $3/[s^2(s+1)] = 3/s^2 - 3/s + 3/(s+1)$; $f(t) = 3t - 3 + 3e^{-t}$.

15 MCQs

MCQ 1. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\}$ equals:

- (a) $s F(s)$
- (b) $F'(s)$
- (c) $F(s)/s$
- (d) $s F(s) - f(0)$

The integral theorem states: one integration in the t -domain divides $F(s)$ by s .

MCQ 2. The formula $\mathcal{L}\left\{\int f(\tau) d\tau\right\} = F(s)/s$ is valid when the lower limit of integration is:

- (a) Any finite constant
- (b) $-\infty$
- (c) 0
- (d) $+\infty$

The derivation requires $g(0) = 0$, which holds only when the lower limit is 0.

MCQ 3. $\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\}$ equals:

- (a) $f(t)/t$
- (b) $f'(t)$
- (c) $t f(t)$
- (d) $\int_0^t f(\tau) d\tau$

Division by s in the s -domain corresponds to integration from 0 to t in the t -domain.

MCQ 4. If f is integrated n times (each from 0 to t), then $\mathcal{L}\{I^n f\}$ equals:

- (a) $s^n F(s)$
- (b) $F(s)/s^n$
- (c) $(-1)^n F^{(n)}(s)$
- (d) $n! F(s)/s$

Each nested integration adds one factor of $1/s$; n integrations give $F(s)/s^n$.

MCQ 5. To solve the Volterra integral equation $f(t) + \int_0^t f(\tau) d\tau = e^t$ using Laplace transforms, the first step yields:

- (a) $F(s)(s - 1) = 1$
 (b) $F(s) + sF(s) = 1/(s - 1)$
 (c) $F(s)\left(1 + \frac{1}{s}\right) = \frac{1}{s - 1}$
 (d) $F(s) \cdot s = e^{-s}$

Take $\mathcal{L}$: $F(s) + F(s)/s = \mathcal{L}\{e^t\} = 1/(s - 1)$, giving $F(s)(1 + 1/s) = 1/(s - 1)$.

16 Common Mistakes Box

Common Mistakes in Laplace Transform of Integrals

Mistake	What Students Do	Correct Approach
Multiplying by s instead of dividing	Write $\mathcal{L}\left\{\int_0^t f\right\} = sF(s)$	Should be $F(s)/s$; multiplying by s gives the derivative, not the integral
Forgetting the lower limit must be 0	Apply $F(s)/s$ for $\int_a^t f$ with $a \neq 0$	The proof requires $g(0) = 0$; if $a \neq 0$, a correction term is needed
Using the formula with an unknown lower limit	Write $\mathcal{L}\left\{\int_\alpha^t f\right\} = F(s)/s$ for arbitrary α	First split: $\int_\alpha^t = \int_0^t - \int_0^\alpha$; transform only the $\int_0^t$ part directly
Not simplifying $F(s)/s$ before inverting	Invert a compound fraction without partial fractions	Always simplify or decompose $F(s)/s$ using partial fractions before inverting
Thinking $\mathcal{L}^{-1}\{F(s)/s\} = f(t)/t$	Confuse the division-by- s rule with the division-by- t rule	$\mathcal{L}^{-1}\{F(s)/s\} = \int_0^t f(\tau) d\tau$ (integrate); division-by- t corresponds to $\int_s^\infty F du$ (Topic 11)

17 Quick Recap

Quick Recap — Topic 09: Laplace Transform of Integrals

- **Main theorem:** $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$ (lower limit must be 0).
- **Proof Method 1 (derivative formula):** Set $g = \int_0^t f$; use $\mathcal{L}\{g'\} = sG(s)$ and $g' = f$ to get $G = F/s$.
- **Proof Method 2 (change of order):** Interchange integration region; inner

integral gives $e^{-s\tau}/s$; result is $F(s)/s$.

- **Inverse form:** $\mathcal{L}^{-1}\{F(s)/s\} = \int_0^t f(\tau) d\tau$ — a “bonus $1/s$ ” means integrate.
- **Repeated integration:** n nested integrals from 0 $\Rightarrow$ divide $F(s)$ by s^n .
- **Volterra equations:** $\mathcal{L}$ converts them to algebra; collect $F(s)$, solve, invert.
- **Duality table:** derivative $\leftrightarrow$ multiply by s ; integral $\leftrightarrow$ divide by s ; $tf(t) \leftrightarrow -F'(s)$; $f(t)/t \leftrightarrow \int_s^\infty F du$.
- **Key restriction:** the formula $F(s)/s$ applies *only* when the lower limit of integration is 0.